

Background: linear, quadratic and nonlinear programming

Outline

Linear programming theory

Duality

The optimality conditions: complementary slackness

Sensitivity analysis and shadow prices

Nonlinear optimization theory

Convexity and quadratic programming

The optimality conditions: the KKT conditions

The sensitivity analysis protocol

What this chapter is about

- ▶ We collect the **theoretical tools** used throughout the course
- ▶ The *theory of linear programming*: duality, optimality conditions, sensitivity analysis
- ▶ Its *nonlinear extension*: convexity, KKT conditions
- ▶ It closes with the **sensitivity protocol** used in every laboratory session
- ▶ It also serves as a quick reference

Linear programming theory

Introduction: Linear programming theory

- ▶ Every maximization LP has a **dual** minimization problem
- ▶ The dual variables are the **shadow prices**: the value of one extra unit of resource
- ▶ **Complementary slackness** tells us when a primal-dual pair is optimal
- ▶ **Sensitivity analysis** tells us how the optimum changes when the data change: the question a decision maker cares about

Sets and data

- ▶ $M = \{1, 2, \dots, m\}$: constraint indices; $N = \{1, 2, \dots, n\}$: variable indices
- ▶ **Data**: costs $c_j \in \mathbb{R}$ ($j \in N$), coefficients $a_{ij} \in \mathbb{R}$, right-hand sides $b_i \in \mathbb{R}$ ($i \in M$)
- ▶ $M_{\leq}, M_{=}, M_{\geq}$ partition M according to the direction of the constraint
- ▶ $N_{\geq 0}, N_{\geq 0}, N_{\leq 0}$ partition N according to the sign of the variable (non-negative, free, non-positive)
- ▶ **Variables**: the x_j of the primal and, in the dual, one π_i for each of the m constraints of the primal

The primal (P) (1/2)

Definition 1: The primal (P), general form: objective and constraints

$$\begin{aligned} (P) \quad & \max \sum_{j \in N} c_j x_j \\ & \text{subject to} \quad \sum_{j \in N} a_{ij} x_j \leq b_i, \quad \forall i \in M_{\leq}, \\ & \quad \quad \quad \sum_{j \in N} a_{ij} x_j = b_i, \quad \forall i \in M_{=}, \\ & \quad \quad \quad \sum_{j \in N} a_{ij} x_j \geq b_i, \quad \forall i \in M_{\geq}, \end{aligned}$$

The primal (P) (2/2)

Definition 2: The primal (P), general form: sign of the variables

$$x_j \geq 0, \quad \forall j \in N_{\geq 0},$$

$$x_j \stackrel{\geq}{\leq} 0, \quad \forall j \in N_{\stackrel{\geq}{\leq} 0},$$

$$x_j \leq 0, \quad \forall j \in N_{\leq 0}.$$

- ▶ Three constraint directions ($\leq$, $=$, $\geq$) and three variable signs (non-negative, free, non-positive)

The dual (D) (1/2)

Definition 3: The dual (D), general form: objective and constraints

$$\begin{aligned}(D) \quad & \min \sum_{i \in M} b_i \pi_i \\ & \text{subject to} \quad \sum_{i \in M} a_{ij} \pi_i \geq c_j, \quad \forall j \in N_{\geq 0}, \\ & \quad \quad \quad \sum_{i \in M} a_{ij} \pi_i = c_j, \quad \forall j \in N_{\leq 0}, \\ & \quad \quad \quad \sum_{i \in M} a_{ij} \pi_i \leq c_j, \quad \forall j \in N_{\leq 0},\end{aligned}$$

The dual (D) (2/2)

Definition 4: The dual (D), general form: sign of the variables

$$\pi_i \geq 0, \quad \forall i \in M_{\leq},$$

$$\pi_i \stackrel{\geq}{\leq} 0, \quad \forall i \in M_{=},$$

$$\pi_i \leq 0, \quad \forall i \in M_{\geq}.$$

- The sign of π_i depends on the direction of constraint i of the primal

The rules of duality (1/2)

Every constraint of the primal generates a dual variable, every variable of the primal a dual constraint:

Primal (max)	Dual (min)
constraint $\leq b_i$ ($i \in M_{\leq}$)	variable $\pi_i \geq 0$
constraint $= b_i$ ($i \in M_{=}$)	variable $\pi_i \geq 0$
constraint $\geq b_i$ ($i \in M_{\geq}$)	variable $\pi_i \leq 0$
variable $x_j \geq 0$ ($j \in N_{\geq 0}$)	constraint $\geq c_j$
variable $x_j \geq 0$ ($j \in N_{\geq 0}$)	constraint $= c_j$
variable $x_j \leq 0$ ($j \in N_{\leq 0}$)	constraint $\leq c_j$

The rules of duality (2/2)

Two quick readings:

- ▶ the *equalities* of the primal give *free* dual variables (and vice versa)
- ▶ in a maximization a resource constraint $\leq$ has $\pi_i \geq 0$ (more resource, more value)

For a *minimization* primal the table is read from right to left

The duality theorems

- ▶ The primal-dual pair is linked by the **two duality theorems**
- ▶ In the first, the *bar* denotes any feasible solution
- ▶ In the second, the *tilde* denotes an optimal solution

Definition 5: Weak duality theorem

For every feasible solution $(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n)$ of the primal (P) and every feasible solution $(\bar{\pi}_1, \bar{\pi}_2, \dots, \bar{\pi}_m)$ of the dual (D) we have

$$\sum_{j \in N} c_j \bar{x}_j \leq \sum_{i \in M} b_i \bar{\pi}_i.$$

Definition 6: Strong duality theorem

The primal (P) has an optimal solution $(\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n)$ if and only if the dual (D) has an optimal solution $(\tilde{\pi}_1, \tilde{\pi}_2, \dots, \tilde{\pi}_m)$, and in that case the two optimal values coincide:

$$\sum_{j \in N} c_j \tilde{x}_j = \sum_{i \in M} b_i \tilde{\pi}_i.$$

What the two theorems say

- ▶ **Weak duality:** every feasible solution of the dual is an *upper bound* for the value of the maximization primal (and every feasible solution of the primal a lower bound for the dual)
- ▶ If two feasible solutions have the same value, they are both optimal
- ▶ **Strong duality:** at the optimum this always happens, the gap closes
- ▶ Existence does not mean uniqueness: there may be more than one optimal solution, but the optimal values coincide

Example: the dual of the 2×2 LP (1/3)

Let us apply the rules to the 2×2 LP, which is exactly in the canonical maximization form (constraints $\leq$, non-negative variables):

$$\begin{array}{ll}\max & 30 x_1 + 50 x_2 \\ \text{subject to} & x_1 + 3 x_2 \leq 90, \\ & 2 x_1 + x_2 \leq 80, \\ & x_1, x_2 \geq 0.\end{array}$$

Example: the dual of the 2×2 LP (2/3)

- ▶ **Step 1 — every constraint generates a dual variable.**
Two constraints $\leq$ in a maximization $\Rightarrow \pi_1, \pi_2 \geq 0$
- ▶ **Step 2 — every variable generates a dual constraint.**
The coefficients are the *columns* of the primal, the right-hand sides are the costs:
 - ▶ from $x_1 \geq 0$ we get $\pi_1 + 2\pi_2 \geq 30$
 - ▶ from $x_2 \geq 0$ we get $3\pi_1 + \pi_2 \geq 50$
- ▶ **Step 3 — the objective swaps the roles:** the right-hand sides (90, 80) become the costs of the dual, which is a *minimization* problem

Example: the dual of the 2×2 LP (3/3)

The dual is:

$$\begin{array}{ll} \min & 90 \pi_1 + 80 \pi_2 \\ \text{subject to} & \pi_1 + 2 \pi_2 \geq 30, \\ & 3 \pi_1 + \pi_2 \geq 50, \\ & \pi_1, \pi_2 \geq 0. \end{array}$$

Example: all the cases in a single LP (1/3)

A *minimization* LP with all the cases of the table: three variables with the three signs and three constraints with the three directions (the table is read from right to left):

$$\begin{array}{llllll} \min & 5x_1 & + & 8x_2 & - & 9x_3 \\ \text{subject to} & x_1 & + & x_2 & & \geq & 30, \\ & x_1 & + & x_2 & - & x_3 & = & 100, \\ & x_1 & - & 2x_2 & & \leq & -20, \\ & x_1 & & & & \geq & 0, \\ & & & x_2 & & \geq & 0, \\ & & & & & x_3 & \leq & 0. \end{array}$$

Example: all the cases in a single LP (2/3)

- ▶ **Step 1 — the dual variables**, one per constraint, with the sign given by the direction:
 - ▶ $\pi_1 \geq 0$ (constraint $\geq$ in a minimization)
 - ▶ $\pi_2 \stackrel{=}{=} 0$ (equality)
 - ▶ $\pi_3 \leq 0$ (constraint $\leq$ in a minimization)
- ▶ **Step 2 — the dual constraints**, one per variable, with the direction given by the sign:
 - ▶ from $x_1 \geq 0$ we get $\pi_1 + \pi_2 + \pi_3 \leq 5$
 - ▶ from $x_2 \stackrel{=}{=} 0$ we get $\pi_1 + \pi_2 - 2\pi_3 = 8$
 - ▶ from $x_3 \leq 0$ we get $-\pi_2 \geq -9$
- ▶ **Step 3 — the objective**: the dual is a maximization,
 $\max 30\pi_1 + 100\pi_2 - 20\pi_3$

Example: all the cases in a single LP (3/3)

The dual is:

$$\begin{array}{llllll} \max & 30 \pi_1 & + & 100 \pi_2 & - & 20 \pi_3 \\ \text{subject to} & \pi_1 & + & \pi_2 & + & \pi_3 & \leq & 5, \\ & \pi_1 & + & \pi_2 & - & 2 \pi_3 & = & 8, \\ & & & - & \pi_2 & & \geq & -9, \\ & \pi_1 & & & & & \geq & 0, \\ & & & \pi_2 & & & \geq & 0, \\ & & & & & \pi_3 & \leq & 0. \end{array}$$

Slacks and reduced costs

To every primal constraint we associate its *slack* $\bar{s}_i$, and to every variable its *reduced cost* $\bar{c}_j$, that is the slack of the corresponding *dual* constraint:

$$\bar{s}_i = \sum_{j \in N} a_{ij} \bar{x}_j - b_i, \quad \forall i \in M, \quad \bar{c}_j = c_j - \sum_{i \in M} a_{ij} \bar{\pi}_i, \quad \forall j \in N$$

- ▶ The slack measures how far constraint i is from being binding
- ▶ The reduced cost measures the coefficient c_j net of the value of the resources consumed by x_j , evaluated at the dual variables

Definition 7: LP optimality conditions (complementary slackness)

A pair of feasible solutions, $(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n)$ for the primal (P) and $(\bar{\pi}_1, \bar{\pi}_2, \dots, \bar{\pi}_m)$ for the dual (D), is optimal for both problems if and only if

$$\bar{\pi}_i \cdot \bar{s}_i = 0, \quad \forall i \in M,$$

$$\bar{x}_j \cdot \bar{c}_j = 0, \quad \forall j \in N.$$

Complementary slackness in words

- ▶ **Slack constraint $\Rightarrow$ dual at zero**

$$\bar{s}_i \neq 0 \Rightarrow \bar{\pi}_i = 0$$

- ▶ **Variable different from zero $\Rightarrow$ null reduced cost**

$$\bar{x}_j \neq 0 \Rightarrow \bar{c}_j = 0$$

- ▶ At the optimum these are *implications*, not equivalences: because of degeneracy an active constraint may have a null dual, and a variable at zero a null reduced cost
- ▶ The verification examples further on check exactly these conditions

Two readings for the laboratory (1/3)

Shadow price $\tilde{\pi}_i$: the marginal value of the right-hand side b_i ,

$$\tilde{\pi}_i = \frac{\partial \tilde{z}}{\partial b_i}$$

- ▶ It holds for b_i within the *validity range of the right-hand side* $[b_i^{\min}, b_i^{\max}]$
- ▶ It is the largest interval in which, varying only b_i , the shadow price remains exactly $\tilde{\pi}_i$ (beyond it, it changes and must be recomputed)
- ▶ Special case: inactive constraint $\Rightarrow \tilde{\pi}_i = 0$

Two readings for the laboratory (2/3)

Reduced cost $\bar{c}_j$ (the slack of the dual constraint):

- ▶ for a variable at zero it says by how much c_j must improve for it to become worth activating
- ▶ it holds for c_j within the *validity range of the coefficient* $[c_j^{\min}, c_j^{\max}]$: the largest interval in which, varying only c_j , the optimal solution does not change
- ▶ for a variable at zero the profitability threshold is precisely one of the two endpoints
- ▶ special case: variable different from zero $\Rightarrow \bar{c}_j = 0$

Two readings for the laboratory (3/3)

Beware of degeneracy:

- ▶ there may exist a variable *at zero with null reduced cost*, even though it is not used
- ▶ hence only the implication holds,

$$\bar{c}_j \neq 0 \Rightarrow \text{variable on one of its bounds}$$

not the converse

The signs of shadow prices (1/4)

The shadow price is always, with a single convention,

$$\tilde{\pi}_i = \frac{\partial \tilde{z}}{\partial b_i}$$

the derivative of the optimal value with respect to the right-hand side of constraint i : by how much $\tilde{z}$ changes if b_i increases by one (small) unit

The sign is always deduced by answering **two questions**

The signs of shadow prices (2/4)

- ▶ **Does increasing b_i enlarge or shrink the feasible region?**
 - ▶ with a constraint $\leq$ it enlarges it (the ceiling is raised)
 - ▶ with a constraint $\geq$ it shrinks it (the requirement grows)
 - ▶ with an equality it merely shifts it
- ▶ **How does a wider region change the optimum?**
 - ▶ it can never worsen it
 - ▶ in a minimization the optimal value decreases or stays the same
 - ▶ in a maximization it increases or stays the same

The signs of shadow prices (3/4)

Combining the two answers:

Direction of the constraint	minimization	maximization
$\leq$ (b_i grows $\Rightarrow$ wider region)	$\tilde{\pi}_i \leq 0$	$\tilde{\pi}_i \geq 0$
$\geq$ (b_i grows $\Rightarrow$ narrower region)	$\tilde{\pi}_i \geq 0$	$\tilde{\pi}_i \leq 0$
$=$	any sign (free dual)	

The signs of shadow prices (4/4)

Two typical readings in a *minimization*:

- ▶ a constraint $\leq$ (a capacity) has $\tilde{\pi}_i \leq 0$: relaxing it reduces the cost, and $|\tilde{\pi}_i|$ is the saving for each extra unit of capacity
- ▶ a constraint $\geq$ (a requirement to be met) has $\tilde{\pi}_i \geq 0$: raising the requirement costs, and $\tilde{\pi}_i$ is the marginal cost

In a *maximization* the readings are reversed

In every case an inactive constraint has $\tilde{\pi}_i = 0$ by complementarity

Example: duality on the 2×2 LP (1/4)

Let us take up again the 2×2 LP and its dual and verify the optimality conditions.

Step 1 — solving we obtain:

$$(\tilde{x}_1, \tilde{x}_2) = (30, 20), \quad \tilde{z} = 1900,$$

$$(\tilde{\pi}_1, \tilde{\pi}_2) = (14, 8), \quad (\bar{c}_1, \bar{c}_2) = (0, 0)$$

with validity ranges

$$[b_1^{\min}, b_1^{\max}] = [40, 240], \quad [b_2^{\min}, b_2^{\max}] = [30, 180],$$

$$[c_1^{\min}, c_1^{\max}] = [50/3, 100], \quad [c_2^{\min}, c_2^{\max}] = [15, 90].$$

Both constraints are active.

Example: duality on the 2×2 LP (2/4)

Step 2 — verification of the shadow prices.

- ▶ One extra unit of b_1 is worth 14, one extra unit of b_2 is worth 8
- ▶ $\tilde{x}_1$ and $\tilde{x}_2$ are different from zero: by complementary slackness their dual constraints must be active ($\bar{c}_1 = \bar{c}_2 = 0$):

$$\tilde{\pi}_1 + 2\tilde{\pi}_2 = 14 + 16 = 30 = c_1, \quad 3\tilde{\pi}_1 + \tilde{\pi}_2 = 42 + 8 = 50 = c_2 \quad \checkmark$$

Example: duality on the 2×2 LP (3/4)

Step 3 — verification of strong duality.

$$\sum_{i \in M} b_i \tilde{\pi}_i = 90 \cdot 14 + 80 \cdot 8 = 1260 + 640 = 1900 = \tilde{z} \quad \checkmark$$

Example: duality on the 2×2 LP (4/4)

Step 4 — verification of the reduced costs. With

$$\bar{c}_j = c_j - \sum_{i \in M} a_{ij} \tilde{\pi}_i:$$

$$\bar{c}_1 = 30 - (1 \cdot 14 + 2 \cdot 8) = 0, \quad \bar{c}_2 = 50 - (3 \cdot 14 + 1 \cdot 8) = 0$$

- ▶ Both null, as for every positive variable at the optimum: the two variables repay exactly the resources they consume
- ▶ Only a variable sitting on one of its bounds can have a reduced cost different from zero

Example: verification on all the cases (1/5)

Let us take up again the “all the cases” LP and its dual: the cases are the same ones that will come back, one at a time, in the application chapters.

Step 1 — solving we obtain:

$$(\tilde{x}_1, \tilde{x}_2, \tilde{x}_3) = (60, 40, 0), \quad \tilde{z} = 620,$$

$$(\tilde{\pi}_1, \tilde{\pi}_2, \tilde{\pi}_3) = (0, 6, -1), \quad (\bar{c}_1, \bar{c}_2, \bar{c}_3) = (0, 0, -3)$$

- ▶ ranges of the right-hand sides: $(-\infty, 100]$, $[30, +\infty)$, $[-200, +\infty)$
- ▶ ranges of the coefficients: $(-\infty, 8]$, $[5, 17]$, $(-\infty, -6]$
- ▶ the second and third constraints are active, with $\tilde{x}_3 = 0$

Example: verification on all the cases (2/5)

Step 2 — shadow prices with the rules of the table.

- ▶ $\tilde{\pi}_1 = 0$, consistent with complementarity: the first constraint is slack ($\tilde{x}_1 + \tilde{x}_2 = 100 > 30$)
- ▶ $\tilde{\pi}_2 = 6$ (free dual variable, here positive)
- ▶ $\tilde{\pi}_3 = -1 \leq 0$, as it must be for a constraint $\leq$ in a minimization

Example: verification on all the cases (3/5)

The dual constraints of x_1 and x_2 (different from zero) must be active:

$$\underbrace{0 + 6 - 1}_{x_1} = 5 = c_1, \quad \underbrace{0 + 6 - 2 \cdot (-1)}_{x_2} = 8 = c_2 \quad \checkmark$$

(x_2 is free: its dual constraint is an equality)

Verification by perturbation:

$$b_2 = 101 \Rightarrow \tilde{z} = 626 (+6), \quad b_3 = -19 \Rightarrow \tilde{z} = 619 (-1) \quad \checkmark$$

Example: verification on all the cases (4/5)

Step 3 — verification of strong duality.

$$\sum_{i \in M} b_i \tilde{\pi}_i = 30 \cdot 0 + 100 \cdot 6 + (-20) \cdot (-1) = 620 = \tilde{z} \quad \checkmark$$

Step 4 — reduced costs, one per sign. With

$$\bar{c}_j = c_j - \sum_{i \in M} a_{ij} \tilde{\pi}_i:$$

$$\bar{c}_1 = 5 - (0 + 6 - 1) = 0, \quad \bar{c}_2 = 8 - (0 + 6 + 2) = 0$$

consistent with $\tilde{x}_1, \tilde{x}_2 \neq 0$

Example: verification on all the cases (5/5)

x_3 equals zero, which for a variable ≤ 0 is its *upper bound*:

$$\bar{c}_3 = -9 - (-1) \cdot 6 = -3$$

- ▶ forcing $x_3 = -1$ would worsen the optimum by 3
- ▶ x_3 would become active only if its coefficient rose to at least -6 (precisely the endpoint $c_3^{\max}$ of the validity range)

Verification:

$$x_3 = -1 \Rightarrow \tilde{z} = 623 (+3) \quad \checkmark$$

Sensitivity analysis

- ▶ Solving an LP gives an optimal solution
- ▶ **Sensitivity analysis** answers the next question, the one that really matters to a decision maker:
how does the optimum change if the data change?
- ▶ In LPs a large part of the answer comes for free together with the solution, in the form of **four numbers** for each constraint and variable

The four numbers (1/2)

- ▶ The **shadow price** $\tilde{\pi}_i$ of a constraint: by how much the optimal value improves if the right-hand side of the constraint increases by one unit
“how much I would pay for one extra unit of that resource”
- ▶ The **validity range of the right-hand side** $[b_i^{\min}, b_i^{\max}]$: where that shadow price remains exact (outside it, the optimization must be redone)

The four numbers (2/2)

- ▶ The **reduced cost** $\bar{c}_j$ of a variable at zero: by how much its objective coefficient must improve for it to become worth activating
- ▶ The **validity range of the objective coefficient** $[c_j^{\min}, c_j^{\max}]$: where the coefficient can vary without the optimal solution changing
- ▶ In the 2×2 LP: c_1 can vary in $[50/3 \approx 16.7, 100]$ and c_2 in $[15, 90]$ without moving the solution $(30, 20)$

Example: shadow prices in practice (1/2)

2×2 LP: optimum 1900 with duals $\tilde{\pi}_1 = 14$ and $\tilde{\pi}_2 = 8$.

Reading. One extra unit of b_1 makes the optimum grow by 14; if obtaining it cost 12, it would be worth it ($14 > 12$).

Verification by perturbation. Re-solving:

$$b_1 = 91 \Rightarrow 1914 = 1900 + 14, \quad b_1 = 100 \Rightarrow 2040 = 1900 + 10 \cdot 14$$

the shadow price is still exact because 100 lies inside the validity range $[40, 240]$.

Example: shadow prices in practice (2/2)

Where it stops holding.

- ▶ At $b_1 = 240$ the first constraint stops being the scarce resource
- ▶ Beyond that, every additional unit is worth less than 14 and everything must be recomputed
- ▶ This is why every shadow price is quoted *together* with its validity range

Inactive constraints.

- ▶ If we had $b_1 = 300$ (constraint inactive at the optimum), its shadow price would be 0
- ▶ A resource left over is worth nothing at the margin

Example: is a third variable worth it? (1/3)

- ▶ In the 2×2 LP both variables are positive at the optimum and their reduced costs equal 0: this is always the case for variables different from zero
- ▶ To see a reduced cost at work we need a variable at zero
- ▶ Let us add to the model a variable $x_3 \geq 0$ with coefficient $c_3 = 20$ in the objective and consumptions $a_{13} = a_{23} = 1$ in the two constraints

Example: is a third variable worth it? (2/3)

Step 1 — valuation with the shadow prices. The resources absorbed by one unit of x_3 are worth, at the shadow prices,

$$1 \cdot 14 + 1 \cdot 8 = 22 > 20 = c_3$$

every unit of x_3 would take resources away from better uses.

Step 2 — the reduced cost. The difference is precisely the reduced cost:

$$\bar{c}_3 = 20 - 22 = -2$$

Solving confirms it: the optimal solution remains $(30, 20, 0)$ with $\tilde{z} = 1900$, x_3 is at zero with $\bar{c}_3 = -2$ and $c_3^{\max} = 22$.

Example: is a third variable worth it? (3/3)

- ▶ The coefficient c_3 would have to rise to at least 22 for it to be worth activating x_3
- ▶ **Step 3 — counter-check.** With $c_3 = 23 > 22$ the solution really does change, and not by a little: the new optimal solution is $(0, 5, 75)$ with value 1975
- ▶ The reduced cost tells us *when* a variable at zero enters the solution, not *how* the solution changes: that has to be recomputed

Summary: Linear programming theory (1/2)

Duality:

- ▶ Every constraint generates a dual variable, every variable a dual constraint
- ▶ Weak duality: the dual gives a bound; strong duality: at the optimum the values coincide

Complementary slackness:

- ▶ $\bar{\pi}_i \cdot \bar{s}_i = 0$ and $\bar{x}_j \cdot \bar{c}_j = 0$
- ▶ Slack constraint $\Rightarrow$ null dual; non-null variable $\Rightarrow$ null reduced cost

Summary: Linear programming theory (2/2)

Sensitivity:

- ▶ Shadow price $\tilde{\pi}_i = \partial \tilde{z} / \partial b_i$, valid within its range $[b_i^{\min}, b_i^{\max}]$
- ▶ The sign is deduced from: does the region widen or narrow?
Min or max?
- ▶ Reduced cost: by how much c_j must improve for a variable at zero to be worth it
- ▶ A number is always quoted *together* with its validity range

Nonlinear optimization theory



Introduction: Nonlinear optimization

- ▶ We extend the tools of LP to **quadratic** and **nonlinear** problems
- ▶ **Convexity** is the property that allows us to certify the optimum
- ▶ The **KKT conditions** generalize complementary slackness
- ▶ The KKT multipliers generalize the shadow prices: we will still have an economic reading

Convex problems

- ▶ A problem $\min f(x_1, x_2, \dots, x_n)$ over a feasible set X is **convex** if
 - ▶ f is convex
 - ▶ X is convex
- ▶ Let us see the property we shall use all the time

Why convexity matters

- ▶ In a convex problem every local minimum is global: optimality can be *certified*
- ▶ In non-convex problems (for instance a pricing problem with a bilinear term $p \cdot q$) a local method guarantees only a local optimum
- ▶ Non-convex QPs can nonetheless be solved to global optimality with dedicated techniques, but the computational cost grows

Quadratic programming

- ▶ A **QP** has objective

$$\frac{1}{2} \sum_i \sum_j q_{ij} x_i x_j + \sum_j c_j x_j$$

and linear constraints

- ▶ It is convex if and only if the matrix $\mathbf{Q} = (q_{ij})$ is positive semidefinite ($\mathbf{Q} \succeq 0$)
- ▶ In this course QPs appear every time a cost or a risk grows more than linearly with the decision

Example: a 2×2 QP (1/3)

A minimization QP with a separable quadratic objective (the second variable “costs” twice as much as the first, at the margin) and a threshold constraint on the sum:

$$\begin{aligned} \min \quad & x_1^2 + 2x_2^2 \\ \text{subject to} \quad & x_1 + x_2 \geq 6, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Here $\mathbf{Q} = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix} \succ 0$: the QP is convex and the optimum found is global.

Example: a 2×2 QP (2/3)

Step 1 — solving we obtain:

$$\tilde{x}_1 = 4, \quad \tilde{x}_2 = 2, \quad \tilde{f} = 24, \quad \lambda = 8$$

Step 2 — verification of the shadow price.

- ▶ The constraint is active ($4 + 2 = 6$): the objective grows with the variables, hence the sum does not exceed what is needed
- ▶ The partial derivatives coincide:

$$2\tilde{x}_1 = 8 = 4\tilde{x}_2 = \lambda \quad \checkmark$$

- ▶ If they were different it would be worth shifting quantity from one variable to the other; the common derivative is precisely the multiplier

Example: a 2×2 QP (3/3)

Step 3 — verification by perturbation.

With right-hand side 7 we obtain $(\tilde{x}_1, \tilde{x}_2) = (14/3, 7/3)$ with

$$\tilde{f} = 32.67 \approx 24 + 8$$

- ▶ On top of that there is the curvature term
- ▶ Here the shadow price is exact only at the margin, not for finite increments

The Lagrangian

- ▶ The optimality conditions of constrained nonlinear problems are known as the **KKT conditions** (after Karush, Kuhn and Tucker)
- ▶ Problem: $\min f(x_1, \dots, x_n)$ subject to
 - ▶ $g_i(x_1, \dots, x_n) \leq 0$ for every $i \in \{1, 2, \dots, m\}$
 - ▶ $h_j(x_1, \dots, x_n) = 0$ for every $j \in \{1, 2, \dots, q\}$
- ▶ The **Lagrangian** is

$$L = f(x_1, \dots, x_n) + \sum_{i=1}^m \lambda_i g_i(x_1, \dots, x_n) + \sum_{j=1}^q \nu_j h_j(x_1, \dots, x_n), \quad \lambda_i \geq 0$$

The KKT conditions

Definition 8: KKT conditions

At a regular optimal point $(\tilde{x}_1, \dots, \tilde{x}_n)$ there exist multipliers $\lambda_i \geq 0$ and ν_j such that

$$\frac{\partial L}{\partial x_k}(\tilde{x}_1, \dots, \tilde{x}_n) = 0 \quad \forall k \in \{1, \dots, n\} \quad (\text{stationarity}),$$

$$\lambda_i g_i(\tilde{x}_1, \dots, \tilde{x}_n) = 0 \quad \forall i \in \{1, \dots, m\} \quad (\text{complementarity}).$$

If the problem is convex, the KKT conditions are also sufficient: a point that satisfies them is a global optimum.

The KKT conditions are complementary slackness, generalized

- ▶ The two parts of the chapter mirror each other: for an LP the KKT conditions reduce *exactly* to complementary slackness
- ▶ The stationarity of the Lagrangian is equivalent to the feasibility of the dual:
 - ▶ the multipliers λ_i are the dual variables π_i
 - ▶ the multipliers of the sign constraints are the reduced costs
- ▶ The complementarity $\lambda_i g_i(\tilde{x}_1, \dots, \tilde{x}_n) = 0$ is the complementary slackness condition $\bar{\pi}_i \cdot \bar{s}_i = 0$
- ▶ Complementary slackness is the KKT conditions in the linear case

Example: the KKT conditions on the QP (1/3)

Let us take up again the QP just solved: $\min x_1^2 + 2x_2^2$ subject to $x_1 + x_2 \geq 6$, $x_1, x_2 \geq 0$.

Step 1. The constraint in standard form:

$$g(x_1, x_2) = 6 - x_1 - x_2 \leq 0$$

Step 2. Stationarity of the Lagrangian

$$L = x_1^2 + 2x_2^2 + \lambda(6 - x_1 - x_2):$$

$$\frac{\partial L}{\partial x_1} = 2x_1 - \lambda = 0, \quad \frac{\partial L}{\partial x_2} = 4x_2 - \lambda = 0 \Rightarrow x_1 = \frac{\lambda}{2}, \quad x_2 = \frac{\lambda}{4}$$

(the constraints $x_1, x_2 \geq 0$ will turn out to be inactive: their multipliers are null by complementarity)

Example: the KKT conditions on the QP (2/3)

Step 3. Complementarity:

- ▶ if the threshold constraint were inactive ($\lambda = 0$) we would have $x_1 = x_2 = 0$, which violates $x_1 + x_2 \geq 6$: impossible
- ▶ hence the constraint is active:

$$\frac{\lambda}{2} + \frac{\lambda}{4} = 6 \Rightarrow \lambda = 8, \quad \tilde{x}_1 = 4, \quad \tilde{x}_2 = 2, \quad \tilde{f} = 24$$

- ▶ the same numbers found by elementary means, now obtained with the general procedure

Example: the KKT conditions on the QP (3/3)

Step 4. Reading of the multiplier:

- ▶ with right-hand side $6 + \varepsilon$ the optimal value grows by about $\lambda\varepsilon = 8\varepsilon$
- ▶ verification: with right-hand side d the solution is $\tilde{x}_1 = 2d/3$, $\tilde{x}_2 = d/3$, hence

$$\tilde{f}(d) = \frac{2}{3}d^2 \Rightarrow \tilde{f}(6 + \varepsilon) = 24 + 8\varepsilon + \frac{2}{3}\varepsilon^2 \quad \checkmark$$

Multipliers and shadow prices

- ▶ The KKT multipliers generalize the shadow prices of the LP: they measure the marginal variation of the optimal value with respect to the right-hand side of the constraint
- ▶ When the duals are not available (general nonlinear models) we estimate the multiplier *by perturbation*:
 - ▶ the right-hand side is increased by ε
 - ▶ the problem is re-optimized
 - ▶ the incremental ratio is computed
- ▶ This numerical verification is itself a recurring exercise

Summary: Nonlinear optimization

Convexity:

- ▶ In a convex problem every local minimum is global
- ▶ A QP is convex if and only if $\mathbf{Q} \succeq 0$

KKT conditions:

- ▶ Stationarity of the Lagrangian and complementarity
- ▶ Sufficient in convex problems
- ▶ In the linear case they are complementary slackness
- ▶ The multipliers are shadow prices; they can also be estimated by perturbation

The sensitivity analysis protocol

Introduction: The sensitivity protocol

- ▶ Sensitivity analysis is **part of the model**, not an appendix
- ▶ In every laboratory session we follow the same **six-step protocol**
- ▶ We close with three exercises on duality, KKT and convexity

The six-step protocol (1/2)

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- | | |
|------------------------------------|--|
| 1. Base scenario | solve, verify feasibility, identify the active constraints |
| 2. One-at-a-time | vary one key parameter over a grid; plot value/decisions |
| 3. Shadow prices and reduced costs | compare the marginal prediction of the dual with a re-optimization after a small perturbation; read the reduced cost $\bar{c}_j$ of every variable at zero: the threshold beyond which it is worth activating it |
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The six-step protocol (2/2)

4. Scenarios	build at least three consistent scenarios (pessimistic, central, optimistic)
5. Trade-off	vary a weight or a threshold to trace a frontier (cost-service, risk-return, cost-emissions)
6. Stability	flag abrupt changes in the optimal mix and fragile estimated parameters

Exercise: duality

- ▶ In the 2×2 LP of the example it is possible to increase b_1 by 10 units by paying 12 per unit. Is it worth it?
- ▶ Up to which unit price is it worth it?
- ▶ Up to which value of b_1 does the shadow price of 14 hold?

Hint: recompute the optimal vertex when the first constraint moves to $90 + \Delta$ and find the Δ beyond which the solution changes structure.

Solution

- ▶ With the first constraint at $90 + \Delta$ the optimal vertex solves $x_1 + 3x_2 = 90 + \Delta$, $2x_1 + x_2 = 80$:

$$\tilde{x}_1 = \frac{150 - \Delta}{5}, \quad \tilde{x}_2 = \frac{100 + 2\Delta}{5}$$

- ▶ $\tilde{x}_1 \geq 0$ as long as $\Delta \leq 150$, that is up to $b_1 = 240$: up to there the shadow price 14 is exact
- ▶ 10 units are worth $10 \cdot 14 = 140$ and cost $10 \cdot 12 = 120$: it is worth it (net gain 20)
- ▶ It is worth it as long as the unit price is below 14

Exercise: KKT

Solve with the KKT conditions:

$$\min (x - 3)^2 + (y - 1)^2 \quad \text{subject to} \quad x + 2y \leq 2, \quad x, y \geq 0$$

- ▶ Say which constraints are active at the optimum
- ▶ Interpret the multipliers

Solution (1/2)

- ▶ The point $(3, 1)$ is not feasible; its projection onto the line $x + 2y = 2$ is $(2.4, -0.2)$, with $y < 0$: hence $y \geq 0$ will be active too
- ▶ Candidate: $x + 2y = 2, y = 0 \Rightarrow (\tilde{x}, \tilde{y}) = (2, 0), \tilde{f} = 1 + 1 = 2$
- ▶ Lagrangian: $L = (x - 3)^2 + (y - 1)^2 + \lambda(x + 2y - 2) - \mu y$

$$\frac{\partial L}{\partial x} = 2(x - 3) + \lambda = -2 + \lambda = 0 \quad \Rightarrow \quad \lambda = 2$$

$$\frac{\partial L}{\partial y} = 2(y - 1) + 2\lambda - \mu = -2 + 4 - \mu = 0 \quad \Rightarrow \quad \mu = 2$$

Solution (2/2)

- ▶ $\lambda = 2 \geq 0$, $\mu = 2 \geq 0$; the constraint $x \geq 0$ is not active ($\tilde{x} = 2$) and has null multiplier: the KKT conditions hold
- ▶ The problem is convex: $(2, 0)$ is the global optimum
- ▶ Active: $x + 2y \leq 2$ and $y \geq 0$
- ▶ $\lambda = 2$: relaxing the constraint to $x + 2y \leq 2 + \varepsilon$ the optimal value decreases by about 2ε
- ▶ $\mu = 2$: if $y \geq -\varepsilon$ were allowed the optimal value would decrease by about 2ε

Exercise: convexity

For which values of the parameter γ is the function

$$f(x_1, x_2) = x_1^2 + \gamma x_1 x_2 + x_2^2$$

convex?

(Write $\mathbf{Q}$ and impose $\mathbf{Q} \succeq 0$.)

Solution

- ▶ With $f = \frac{1}{2} \sum_i \sum_j q_{ij} x_i x_j$:

$$\mathbf{Q} = \begin{pmatrix} 2 & \gamma \\ \gamma & 2 \end{pmatrix}$$

- ▶ A symmetric 2×2 matrix is positive semidefinite if and only if the diagonal entries and the determinant are non-negative:

$$2 \geq 0, \quad \det \mathbf{Q} = 4 - \gamma^2 \geq 0$$

- ▶ Hence f is convex if and only if $-2 \leq \gamma \leq 2$

Summary: The sensitivity protocol

- ▶ Sensitivity is part of the model
- ▶ Six steps: base scenario, one-at-a-time, shadow prices and reduced costs, scenarios, trade-off, stability
- ▶ Every marginal prediction must be compared with a re-optimization

Final summary (1/2)

Linear programming:

- ▶ Primal and dual, linked by weak and strong duality
- ▶ Complementary slackness as optimality conditions
- ▶ Shadow prices, reduced costs and validity ranges

Nonlinear optimization:

- ▶ Convexity: every local minimum is global
- ▶ KKT conditions: complementary slackness, generalized

Final summary (2/2)

In the laboratory:

- ▶ Shadow prices (and multipliers) are verified by perturbation
- ▶ Every shadow price is quoted with its validity range
- ▶ Every laboratory session follows the six-step sensitivity protocol